PROCESS OR MATERIALS REVIEW REQUEST FORM

SNF Stanford Nanofabrication Facility

Request Title: N2 Annealing in clean Thermco

Requestor: Payton Broaddus **Date: 2019/3/15**

Non-SNF staff, please complete the following:

Badger ID: broaddus

Company name/PI: Harris

Send completed form to: snf-promcommittee@lists.stanford.edu

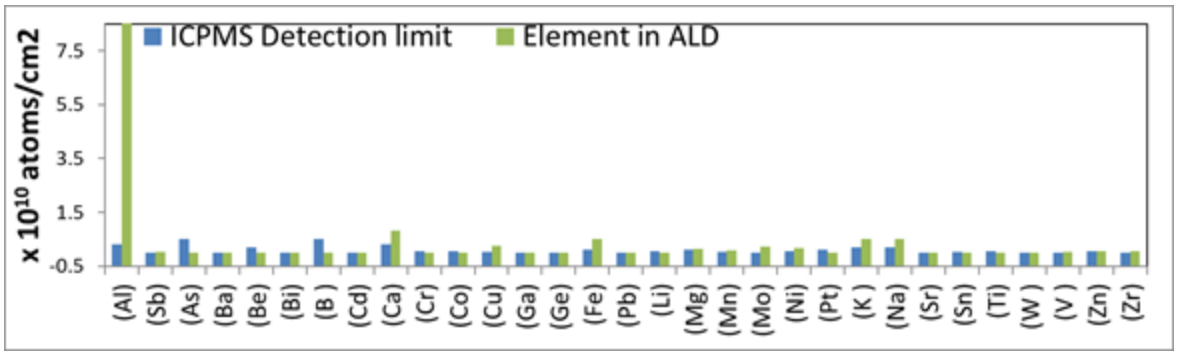
1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the Stanford Chemical Storage Groups and the Stanford Chemical Safety Data Base sections on this website to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

What wafer?
New Si wafer
What cleaning done?
No cleaning. Wafer from new box or previous open vacuum bagged box
What processing done?
ALD process at 1000 clean room
What Deposited on Wafer?
Thickness of two Al2O3 films: 100nm and 1um (a week of ALD)
What cleanliness of tool?
Semiconductor grade cleaned. Contamination level is below 1E10 for 31 metals (see below). Al is higher, since film deposited is Al2O3.
What dicing done?
Wafer cleaved by diamond scriber at 100 clean room before loading to ALD tool
What cleanliness of dicing?
Cleaving by diamond scriber done at 100 clean room
What packaging/cleanliness?
After process wafer transfer in the box and clean room ziploc
(See attached ICPMS output)

Have the following tools, other than ALD chamber, used to transfer, load, and dice wafers/dices ever been used to handle metal?

It is a standalone single chamber. Never use metal handler. This includes teflon coated tweezers used to pick up the dies.

ICPMS of ALD tool:



2. **Vendor/manufacturer info:** address and phone number, website URL

Applied Materials.

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

I want to do a N2 Anneal @ 1100C for (1 and 10 hrs) of three diced samples (2 with Al2O3 ALD coating, one without) and then characterize any phase shift of the alumina with XRD, GIXRD, raman, and SEM/EDS, most importantly GIXRD. Since we don't want any surface metal contamination for SEM/EDS and Raman, we want to use a clean N2 (thermco123) tool rather than a contaminated one with unknown contaminates (thermco4).

The only tool two tools used to processes these dies previously is a diamond dicer at Applied Materials and the Applied Materials ALD tool, which deposited 1um (a week long process) and 100nm of ALD Al2O3.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.) Make sure to include wet benches. Please note that f the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed

into a "clean" tool, the material should MOS grade or better.

1um Al2O3 ALD, 100nm Al2O3 ALD, 0nm Al2O3 ALD: Unpack with scissors, NO RCA clean, load in thermco1 with clean wand:

-1 hr 1100C N2 Anneal (thermco)

-10 hr 1100C N2 Anneal (thermco)

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

6 Si Dies, x2 1um Al2O3 ALD, x2 100nm Al2O3 AlD, x2 0nm.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). Storage groups A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while storage groups C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

N/A

7. **Disposal**: How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: